

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL A-LEVEL PSYCHOLOGY

Unit 3 Advanced Topics and Research Methods 2

Wednesday 14 June 2023 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a calculator, which you are expected to use where appropriate.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- All working must be shown.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 90.

For Examiner's Use	
Section	Mark
A	
B	
C	
TOTAL	



Section A: Psychology of Sleep

Answer **all** questions in the spaces provided.

0 1

Briefly outline **one** characteristic of narcolepsy.

[2 marks]

0 2

Briefly evaluate research into narcolepsy.

[4 marks]



0	3
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Two students are talking about animal behaviours.

“Animal sleeping habits are fascinating,” says Olga. “Bears sleep through the entire winter while other animals, like hamsters, sleep all day but are awake and active at night.”

“And I’ve always wondered why squirrels or cats may sleep up to 20 hours a day!” replies Anya.

Using your knowledge of **at least one** evolutionary explanation for the functions of sleep, explain the conversation between Olga and Anya.

[4 marks]

Turn over for the next question

Turn over ►



Discuss the effects of endogenous pacemakers **and** exogenous zeitgebers on the sleep/wake cycle.

[illegible]

[illegible]

Turn over ►



Section B: SchizophreniaAnswer **all** questions in the spaces provided.Only **one** answer per question is allowed.

For each question completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.

**0 5**

What is meant by co-morbidity?

[2 marks]

0 6

Aisha and her twin brother Zachary are aged 18. They both show symptoms of schizophrenia, such as hallucinations and delusions. Zachary also shows signs of avolition and speech poverty. Aisha decides to talk to her friends about her symptoms but Zachary refuses to talk to his friends.

Referring to gender bias, explain why Zachary is more likely to be diagnosed with schizophrenia than Aisha.

[3 marks]



0	9
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Explain how anti-psychotic drugs treat schizophrenia.

[3 marks]

1	0
---	---

Explain **one** limitation of using anti-psychotic drugs to treat schizophrenia.

[3 marks]



[12 marks]

[illegible]

[illegible]

Section C: Research Methods 2

Answer **all** questions in the spaces provided.

As part of a school project, college students decided to test if there was a difference in how quickly people can read text printed on yellow paper or green paper. They used a repeated measures design with a group of 20 psychology students. All 20 were asked to read two similar texts, and the conditions were counterbalanced as follows:

- 10 participants read Text 1 printed on yellow paper; they rested for 30 minutes, then read Text 2 printed on green paper
- the other 10 participants read Text 1 on green paper; they rested for 30 minutes, then read Text 2 printed on yellow paper.

The students collected their data using a stopwatch. They then compared the speed of reading of each participant for both the yellow paper and the green paper texts.

1	2
---	---

The students used a repeated measures design. Explain **one** advantage of using a repeated measures design in this study.

[2 marks]

Turn over for the next question

Turn over ►



1	3
---	---

The students collected the data using an interval level of measurement.

Explain what is meant by an interval level of measurement in this study.

[2 marks]

1	4
---	---

The students used a related t-test to check if the results were significant.

1	4
---	---

1

Apart from the interval level of measurement, outline **two** other reasons why the related t-test was appropriate for this study.

[2 marks]



1 4 . 2 The students calculated the t score for the related t-test, $t = 2.899$

Table 1 Critical values for the related t-test

	Levels of significance for a two-tailed test			
	0.1	0.05	0.02	0.01
df	/	/	/	/
18	1.734	2.101	2.552	2.878
19	1.729	2.093	2.539	2.861
20	1.725	2.086	2.528	2.845
21	1.721	2.080	2.518	2.831

The calculated value must be equal to or greater than the critical value in the table for the significance to be shown.

Degrees of freedom (df) for the related t-test can be calculated by using the formula $df = N - 1$

Using the information in **Table 1**, explain whether the results of the study are significant or not. Justify your answer.

[3 marks]

Turn over for the next question

Turn over ►



1	5
---	---

The students were worried that they might have made a Type I error in their study.

1	5	.	1
---	---	---	---

What is a Type I error?

[2 marks]

1	5	.	2
---	---	---	---

Using the critical values from **Table 1**, explain why the students were unlikely to have made a Type I error.

[2 marks]



1 6

After measuring the speed of reading, the students designed a 20-item memory test to investigate if the colour of the paper might also affect the ability to remember printed information.

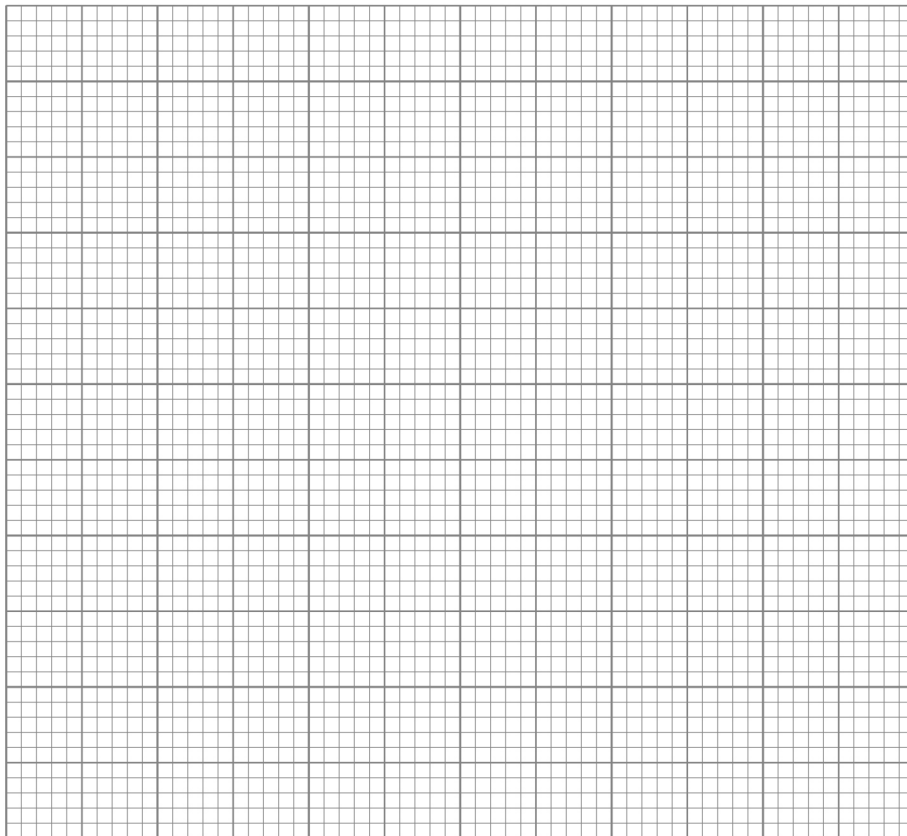
Below are the mean values of correctly recalled memory items from each of the texts.

Table 2 Mean values for memory recall for green paper and yellow paper

Green paper (out of 20 items)	Yellow paper (out of 20 items)
12	16

Using the graph paper below, draw an appropriate display of the data in **Table 2**. Label your display fully, including the title.

[5 marks]



Turn over ►



1 7

The students decided to ask the participants their opinions after they had taken part in the memory test. They did this by using a questionnaire.

Outline **one** advantage of using a questionnaire rather than an interview.

[2 marks]

1 8

Write a closed question that could be used to collect participants' opinions about the difficulty of the memory test.

[1 mark]

1 9

Many psychologists use case studies as a research method.

Outline and evaluate the use of case studies in helping psychologists to understand human behaviour.

[9 marks]



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30



There are no questions printed on this page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**



[illegible]

